Dr. K V Sriharsha
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LinkedIn | Google Scholar

PROFILE

AI and Computer Vision professional with 14+ years of experience in industrial R&D;, AI-driven automation, academia, and research. Expertise in Computer Vision, Deep Learning, Machine Learning, Industrial AI, Embedded Vision, and Digital Transformation. Proven experience in AI-based inspection systems, defect localization, surveillance systems, and industrial automation.

CORE SKILL SET

Programming & Frameworks: Python, C, MATLAB, J2SE, OpenCV, TensorFlow, PyTorch, Keras, Scikit-learn, TensorRT, NumPy, SciPy, Matplotlib

AI & Computer Vision: Object Detection & Tracking, OCR, Image Segmentation, Face Anti-Spoofing, 3D Vision, Depth Estimation, Vision Transformers, Embedded AI, Industrial Defect Detection

3D Imaging & Sensors: Stereo Vision, Structured Light Imaging, ToF Imaging, LiDAR, Intel RealSense, Kinect

Embedded & Hardware: Raspberry Pi, Arduino, NVIDIA Jetson Nano, CMOS Imaging, Vision Sensors

Protocols & Systems: UART, SPI, I2C, USB, HDMI, Bluetooth, ZigBee, MQTT, Wi■Fi, Ethernet

Industrial Skills: AI Project Management, Predictive Maintenance, MLOps, CI/CD, AI Deployment, Technical Proposal Writing, Product Commercialization

PROFESSIONAL EXPERIENCE

OLA Electric Technologies Pvt Ltd – Bengaluru
AI Project Lead – Automation (R&D;) | *Dec 2023 – Present*
• Leading AI-driven automation initiatives for manufacturing systems.
• Deploying computer vision models for industrial inspection and quality monitoring.
• Coordinating engineering and data science teams for AI implementation.

Armatec Global – Sydney (Remote/Onsite)
AI/CV Engineer (R&D;) | *Sep 2023 – Nov 2023*
• Delivered AI consulting for industrial computer vision solutions.
• Coordinated with international stakeholders for deployment activities.

Matdun Labs India Pvt Ltd – Hyderabad
AI/CV Engineer (R&D;) | *Dec 2021 – May 2023*
• Delivered AI-based computer vision applications for startups and industrial clients.
• Built datasets and trained deep learning models for defect localization and inspection.

National Institute of Technology (NIT), Trichy

Senior Research Fellow – Computer Vision | Aug 2015 – Feb 2021
• Conducted research in depth sensing, computational photography, and computer vision.
• Published papers in SCI/SCIE journals and IEEE conferences.
• Mentored PG/PhD students in AI, ML, and Deep Learning.

EDUCATION

PhD – Computer Vision & AI, NIT Trichy (2021)
M.Tech – Information Technology, SRKR Engineering College (2008)
MSc – Electronics, GVP College for PG Courses (2006)
BSc – Electronics, Gayatri Vidya Parishad (2004)

PhD Thesis: "Depth Sensing Mechanism using Optical Imaging with Inherent Machine Learning and Deep Learning Techniques"

SELECT INDUSTRIAL PROJECTS

• Vision-Based Defect Detection in Battery Manufacturing
• AI-Based Weld Inspection System for Automobile Manufacturing
• Face Anti-Spoofing using Structured Light & ToF Imaging
• Factory Surveillance & Proximity Monitoring Systems
• Embedded Vision-Based Facial Recognition Systems

RESEARCH CONTRIBUTIONS

• Developed non-triangulation-based monocular depth estimation models using aperture, magnification, and blur cues for low-cost RGB imaging systems.

• Integrated computational photography concepts with AI-driven sensing systems for scalable real-world deployment.

• Designed single-tier visual sensor network architectures and real-time face recognition pipelines for edge-based systems.

• Developed Kalman filter-based object tracking and mean-shift integrated tracking frameworks for dynamic scene analysis.

• Proposed AI + geometry hybrid learning frameworks for robust depth inference and reduced data dependency.

• Contributed to industrial AI systems for EV manufacturing, intelligent inspection, and AI-driven automation deployments.

• Developed datasets and AI pipelines for industrial defect localization, battery manufacturing inspection, and surveillance applications.

• Worked on interdisciplinary AI applications spanning computational imaging, industrial AI, healthcare AI, and smart sensing systems.

PATENTS

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1. A Dual Reference Axis Model for Target Localization in 3D Space using AI Based Techniques – Indian Patent Granted (2025)

2. AI Based Device for Skin Cancer Detection – Design Patent Granted (2025)

3. Non Invasive Glucose Monitoring System and Method Thereof – Indian Patent Published (2026)

AWARDS & ACHIEVEMENTS

- Best Paper Award – IEEE International Conference on Computational Intelligence & Computing Research
- Visvesvaraya PhD Fellowship Awardee – MeitY, Government of India
- Silver Partner Faculty – Infosys Campus Connect
- Gold Medal – Best Outgoing Student (BSc)

CERTIFICATIONS

- Deep Learning Specialization – DeepLearning.AI (Coursera)
- Machine Learning Crash Course – Google Developers
- Deep Learning Workshops – IEEE, IIT Roorkee, NIT Trichy